Visualization and Simulation of Transportation Data

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Introduction and Background

To support infrastructure planning and policy decisions regarding transportation in the United States, large-scale datasets are collected to better understand both travel demand and driver behavior. Two widely used datasets in transportation engineering are the National Household Travel Survey (NHTS) and the Next Generation Simulation (NGSIM) dataset.

The NHTS dataset offers insights into travel behaviors at the national scale, including trip frequency, timing, and individual vehicle characteristics. In contrast, the NGSIM dataset provides detailed vehicle trajectory data, enabling comprehensive analysis of driver behavior, vehicle interactions, and traffic flow dynamics.

This project aims to utilize Python-based tools to process, visualize, and analyze these datasets in a transparent and reproducible manner. The primary objectives are to identify key transportation trends through statistical visualizations and to evaluate vehicle-following behavior using a simulation study based on the Intelligent Driver Model (IDM).

The results of this analysis are intended to inform transportation planning decisions by offering both broad insights into travel behavior and detailed perspectives on individual vehicle interactions.

Definitions

These definitions establish the foundational concepts used throughout the analysis and simulation presented in this report.

National Household Travel Survey:

The National Household Travel Survey (NHTS) is a large-scale dataset collected by the Federal Highway Administration (FHWA) that captures travel behavior across the United States. It includes information on trip frequency, travel timing, vehicle characteristics, and geographic distribution based on household-reported data.

The dataset is primarily used to analyze transportation demand patterns at a national and/or regional level, allowing for better infrastructure planning, policy development, and travel trend studies.

Next Generation Simulation:

The Next Generation Simulation (NGSIM) dataset is a high-resolution traffic dataset developed by the FHWA to study detailed vehicle interactions and driving behavior. It provides trajectory-level data collected at brief time intervals (typically 0.1 seconds), including vehicle position, speed, and acceleration.

This dataset is commonly used for analyzing car-following behavior, traffic flow dynamics, and the development and validation of traffic simulation models.

Intelligent Driver Model:

The Intelligent Driver Model (IDM), developed by Treiber, Hennecke, and Helbing (2000), is a widely used physics-based car-following model used to simulate the behavior of a following vehicle in response to a leading vehicle. The model calculates acceleration as a function of the vehicle's current speed, desired speed, spacing (headway), and relative speed between vehicles.

IDM is widely used in transportation engineering to model traffic flow, evaluate driver behavior, and support simulation-based analysis of vehicle interactions.

Scope of Work and Objectives

This project consists of the development of a Python-based system to analyze, visualize, and simulate transportation data using the National Household Travel Survey (NHTS) and the Next Generation Simulation (NGSIM) datasets. The scope includes data processing, visualization, behavioral analysis, and simulation modeling to support transportation engineering insights.

The work completed on this project is divided into the following tasks:

Data Review and Variable Selection

The NHTS and NGSIM datasets were reviewed to identify variables that showcase both transportation characteristics and vehicle interactions/behaviors.

For the NHTS dataset, variables related to travel timing, vehicle characteristics, and geographic distribution were selected. For the NGSIM dataset, trajectory-based variables were selected to analyze vehicle motion and interactions.

Derived variables, including space headway and relative speed, were computed to support analysis of vehicle-following behavior.

Data Cleaning and Processing

Both datasets were processed to ensure usability and consistency. This included handling missing or invalid values, converting data types, and filtering data for analysis.

Numeric conversions were applied where necessary, and non-usable data entries were removed or corrected. These steps ensured that all subsequent analysis and visualization outputs were based on valid and reliable data.

Visualization Development

A series of visualizations were developed to represent transportation trends and behaviors. These included: Bar charts, Histograms, Boxplots and Time-series plots.

All visualizations were created using Python libraries and were formatted with appropriate labeling, scaling, and layout adjustments to ensure clarity and readability.

Intelligent Driver Model Simulation

An Intelligent Driver Model (IDM) simulation was implemented to analyze vehicle-following behavior. The model was executed using a time-step approach, where acceleration, velocity, and position were updated iteratively.

Model inputs included desired velocity, acceleration parameters, spacing, and relative speed. The simulation outputs included vehicle speed profiles, spacing behavior, and trajectory comparisons.

Code and Visualization Validation

All visualizations and outputs were reviewed to ensure consistency. Comparisons between observed and simulated data were performed to evaluate model performance.

The review consisted of:

- Correct representation of selected data
- Confirm logic consistency in trends or distributions.
- Ensure IDM simulation reflects realistic/probable vehicle behaviors.

Prepare Deliverables

The following documents and documentation should be developed and submitted with the final project/report:

Gantt Chart

A project schedule identifying sequencing, dependencies, duration, and milestones.

Scope of work

A formal/professional document outlining project objectives, required tasks, and deliverables.

GitHub Repository

A centralized, shareable repository containing the Python program, a README document explaining the use of the program, and any supporting documentation.

Timesheet

A clearly defined engineering timesheet detailing person and task contributions.

Time Tracking and Evaluation

Time spent on individual tasks throughout the project were captured and documented. This information was used in an engineering timesheet detailing person and task contributions.

The collected data will be used to verify efficiency of performing data analysis in a python environment compared to traditional Excel-based methods.

Methods

Data Sources and Importing

Two datasets were utilized in the analysis: the National Household Travel Survey (NHTS) and the Next Generation Simulation (NGSIM). The NHTS captures travel behavior at the household level, while the NGSIM offers high-resolution trajectory data collected at 0.1-second intervals on urban freeways.

Both NGSIM and NHTS datasets were imported into a Python environment using the Pandas library and the standard '.read_csv()' function. The 'vehicle_age' column was converted to a numeric format using Pandas 'pd.to_numeric()' with 'errors=coerce' to replace invalid entries with NaN values. This approach facilitated the subsequent removal of invalid data points during later stages of analysis.

Data Handling and Processing

Initial processing was performed to ensure reproducibility across each analysis. Mainly, variable formats were standardized, and missing variables were removed/replaced. For the NHTS data, missing values within the “vehicle_age” were removed by appending “.dropna()” before plotting. When performing categorical analysis, counts were computed using “.value_counts()” and reindexed to force consistent category ordering.

In the NGSIM dataset, the focus was the trajectory identifier and time. The vehicle trajectory data was narrowed down based on the desired trajectory identifier and then filtered again by time. This maintained proper chronological order of the data for each trajectory identifier. In the case of this project, trajectories 12 and 13 were used. Any variables derived were calculated directly from the filtered data. There were two distinct instances in this project:

- Space Headway = “leader_position” – “follower_position”
- Relative Speed = “leader_speed” – “follower_speed”

These derivations were appended to the dataset ass new columns for use in any time-series analysis and visualizations.

Variable Selection

Variables used were selected based on relevance to transportation planning, performance, and interactions between vehicles.

NHTS Dataset

Three variables were selected from this dataset:

- travel_day: Data on when vehicles are on the road. This data allows for the evaluation of any variations in trip frequency across the week analyzed. This applies to the categorical structure/relevance to road demand over time.

- vehicle_age: Datapoints on the age of each vehicle within the assessed area. This data can be used to assess the distribution across the nation of vehicle age. These results apply to emissions, future infrastructure planning, and vehicle build quality.
- census_region: Geolocation of each datapoint. This data allows the evaluation of regional differences in vehicle age distributions which can reflect geographic and economic factors.

NGSIM Dataset

Five variables were primarily selected from this dataset:

- Time: This was used as the independent variable for all of the Time-Series plots.
- leader_speed and follower_speed: These were used to analyze leader-follower speed behaviors.
- leader_position and follower_position: These were used to compute spacing and confirm simulation outputs.

Trajectory identifiers 12 and 13 were arbitrarily selected to represent different leader-follower scenarios within the NGSIM dataset.

Plot Construction

All plots were generated using seaborn and matplotlib. A consistent theme "whitegrid" was applied using "sns.set_theme(style="whitegrid")".

Plot 1 – Bar Chart (Trips by Day of Week)

Trip counts were calculated using ".value_counts()" on the "travel_day" column. The count provided was indexed to force a Monday-Sunday order. If data was missing in this category, it was filled using ".fillna(0)" to ensure complete data. The processed data from this was plotted using "sns.barplot()". Axis labels and titles were applied, and to best display x axis headers, the axis was rotated by 20 degrees.

Plot 2 – Histogram (Vehicle Age Distribution)

Rows with missing "vehicle_age" values were removed by ".dropna()". A histogram via "sns.histplot()" was created containing 24 bins. 24 was selected to provide a clear resolution across the observed range while still being readable.

Plot 3 – Boxplot (Vehicle Age by Census Region)

The dataset was grouped by "census_region" and a boxplot was created using "sns.boxplot()" with the dependent variable of "vehicle_age". Regions were indexed to force an order of Northeast, Midwest, South, and West to maintain consistency with the Census Bureau.

Plot 4 – Time-Series (Leader and Follower Speed)

For trajectory identifiers chosen, data was filtered and sorted by time. The leader and follower speeds were plotted as line graphs using "plt.plot()". A legend, axis lables and gridlines were

added to allow clear interpretation and comparison between vehicle behaviors. Each trajectory was plotted as separate figures.

Plot 5 – Dual-Axis Time-Series (Space Headway and Relative Speed)

Derived variables for both space headway and relative speed were computed for trajectory 13. A dual-axis plot was created using "ax1.twinx()", which allows both variables to share the same time axis while maintaining separate y-axis scales. Space headway was plotted on the left axis (solid line), and relative speed was plotted on the right (dashed). Axis labels and tick colors were color-coded to match their respective data series. A combined legend was added by merging handles from both axes.

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Intelligent Driver Model (IDM) Simulation

The Intelligent Driver Model (IDM) is a physics-based, leader-follower model that calculates the following vehicle's acceleration as a function of its current speed. It also finds the gap to the lead vehicle, and the speed differential between the two vehicles.
The governing acceleration equation is Equation 1:

EQUATION 1: IDM ACCELERATION EQUATION

$$\frac{dv}{dt} = a\left[1 - \left(\frac{v}{v_0}\right)^{\delta} - \left(\frac{s^*(v, v\Delta)}{s}\right)^2\right]$$

Where Equation 2 is the desired dynamic spacing.

EQUATION 2: SUPPORTING IDM EQUATION

$$s^*(v, v\Delta) = s_0 + v \cdot T + \frac{v \cdot \Delta v}{2\sqrt{a \cdot b}}$$

The IDM acceleration function was evaluated at each time-step using both observed trajectory data and the simulated follower data. The primary inputs to the model at each time step were the follower speed **v**, the space headway **s**, and the relative speed **Δv**. The follower speed **v** was obtained from the previous simulation step.

The space headway was computed as the difference between the observed leader position and the simulated follower position. Whereas the relative speed was calculated as the difference between the observed leader's speed and the simulated follower's speed.

In addition to these time-dependent inputs, model parameters were defined to represent a driver's behavior. These included the desired free-flow speed $\mathbf{v_0}$, maximum acceleration $\mathbf{a_{max}}$, comfortable deceleration **b**, desired time headway **T**, minimum spacing $\mathbf{s_0}$, and acceleration exponent **δ**. These parameters remained fixed throughout the simulation and controlled how the follower vehicle responded to changing traffic conditions.

The following parameter values were used, consistent with typical urban freeway driving conditions:

TABLE 1: PARAMETERS OF IDM

Parameter	Value	Description
a_{max}	1.2 m/s^2	Maximum acceleration
b	1.8 m/s^2	Comfortable deceleration
δ	4	Acceleration exponent
t_h	1.5 s	Desired time headway
s_o	2.0 m	Minimum gap at standstill
v_o	30 m/s	Desired free-flow speed
Vehicle Length	5.0 m	Used to compute net gap

The simulation was initialized using the observed follower's position and speed at the first of the time-step for trajectory 13. The observed leader trajectory was used as input. The time step "dt" was computed as the median of the differences in the Time column to account for any irregular sampling. The simulation was continued by using Euler integration for velocity (Equation 3) and position (Equation 4):

EQUATION 3: VELOCITY EULER EQUATION

$$v_i + 1 = max(v_i + ai \cdot \Delta t, 0)$$

EQUATION 4: POSITION EULER EQUATION

$$x_i + 1 = x_i + v_i \cdot \Delta t + \frac{1}{2} a_i \cdot \Delta t^2$$

Speed was restricted to stay positive (≥0) to prevent unnatural negative values. The gap was also restricted to 0.1 m to prevent division by zero in outlier cases. Simulated speed and position were compared against observed follower data using Equation 5, the Root Mean Squared Error (RMSE):

EQUATION 5: RMSE EQUATION

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2}$$

Speed RMSE is reported in m/s and position RMSE in m. A lower RMSE value indicates closer correlation between the model and observations, with an RMSE of zero being "perfect

representation". The simulation therefore represents a hybrid approach, where observed leader behavior drives the system while follower behavior is generated through the IDM formulation.

Known Shortfalls

The analysis and simulation conducted in this study are subject to several limitations. The NHTS dataset relies on survey-based data, which may introduce reporting inconsistencies or bias in the representation of travel behavior. Additionally, data cleaning and filtering steps may have removed certain edge cases, potentially limiting the dataset's completeness.

The Intelligent Driver Model (IDM) assumes consistent driver behavior and does not account for sudden or erratic responses. As a result, the model produces softer acceleration and deceleration patterns than those observed in the data, particularly during rapid traffic changes.

Despite these limitations, the overall trends and simulation results provide a reasonable representation of real-world transportation behavior.

Results

Travel Behavior Trends (NHTS)

The distribution of trips shown in Figure 1 across days of the week shows little to no singular-day travel demand. Weekdays, particularly Monday and Sunday, record the highest number of trips, while Saturday shows a relatively sharp decrease in travel activity. This pattern suggests that travel demand is influenced by both work-related commuting and leisure travel.

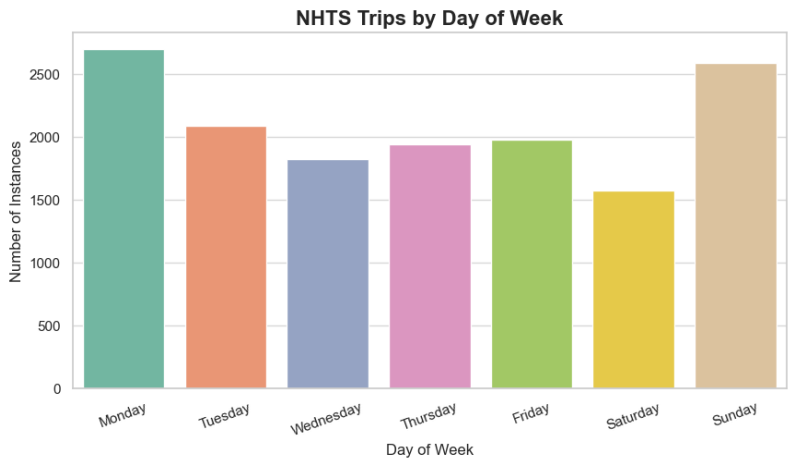

FIGURE 1: BAR CHART OF TRIPS BY DAY OF WEEK

The elevated number of weekday trips reflects expected commuter activity, as individuals travel for employment, education, and routine obligations. The reduced travel volume on Saturday likely indicates a drop in routine or structured trips. The increase on Sunday, though, may suggest recreational travel or errand trips, like shopping, before the workweek.

From a transportation planning perspective, these trends highlight the importance of prioritizing infrastructure capacity and traffic management strategies during peak periods, while recognizing different travel patterns on off days, typically weekends.

Vehicle Age Distribution (NHTS)

The histogram in Figure 2 reveals a skewed distribution, with most vehicles concentrated in the younger age range (approximately 0-10 years) and a decreasing frequency as vehicle age increases. A quite small number of significantly older vehicles (20+ years) are also present.

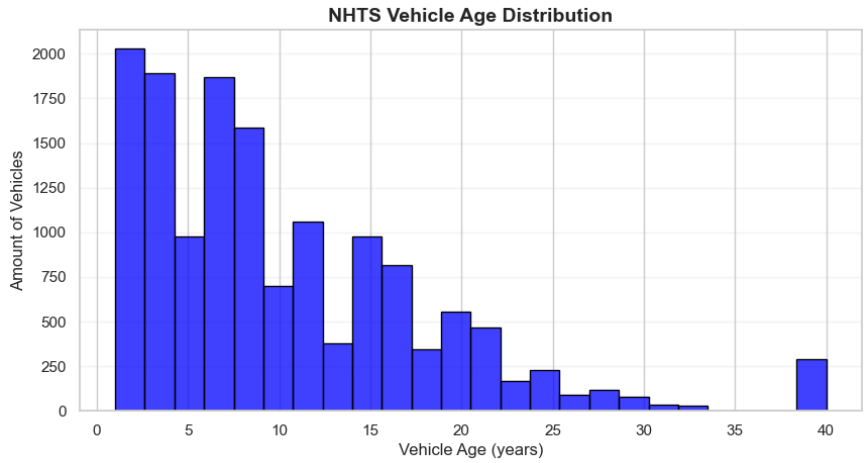

FIGURE 2: HISTOGRAM OF VEHICLE AGE ACROSS DATASET

This distribution shows that while most vehicles on the road are relatively modern, a substantial number are older. This has impacts across multiple facets, such as emissions and safety standards. Older vehicles are also typically associated with reduced fuel efficiency and fewer sophisticated safety features.

For transportation planning purposes, this suggests that policies related to emissions reduction, vehicle inspection programs, and general safety decisions are still relevant points of discussion.

Regional Differences in Vehicle Age (NHTS)

The boxplot in Figure 3 below, comparing vehicle age across census regions, shows noticeable variation in both median values and overall spread. The Midwest and South exhibit slightly higher median vehicle ages than the Northeast and West, and display a larger distribution.

This suggests that vehicles in these regions tend to remain in service longer, which may be affected by economic conditions and typical workforce requirements. Regions with more rural characteristics often experience longer vehicle lifespans due to lower replacement requirements and higher use of private vehicles.

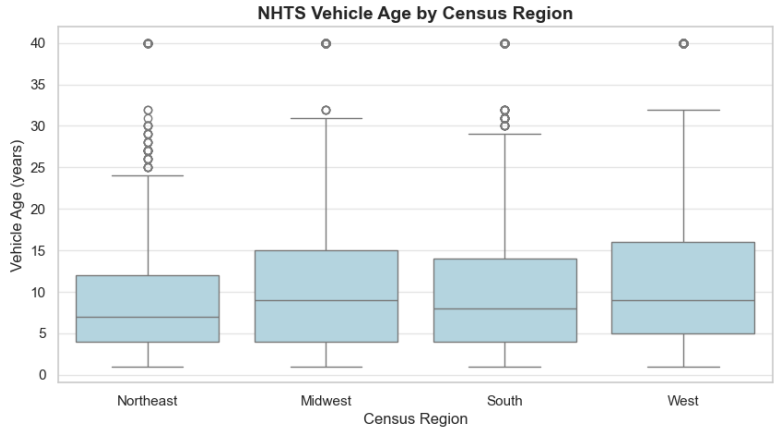

FIGURE 3: BOXPLOT OF VEHICLE AGE IN EACH CENSUS REGION

These regional differences are important for infrastructure planning, as older vehicles may require different maintenance, especially in terms of safety.

Vehicle Interaction and Behaviors (NGSIM)

The time-series plots for trajectories 12 (Figure 4) and 13 (Figure 5) demonstrate clear leader-follower relationships in vehicle speed. In both cases, the follower vehicle generally mirrors the leader's speed trends but with a slight delay and rebounding effect.

This lag response reflects realistic driver behavior, where acceleration and deceleration decisions are made based on perceived changes in the lead vehicle's motion. The plots also show periods of rapid deceleration followed by gradual recovery, which is characteristic of stop-and-go traffic conditions. These dynamics are significant because they contribute to traffic instability and congestion propagation.

Although both trajectories exhibit similar overall trends, trajectory 12 shows more pronounced fluctuations and larger deviations between the leader and follower speeds in quick deceleration and recovery. On the other hand, trajectory 13 displays a smoother and more consistent response, with the follower more closely tracking the leader throughout the entire period.

These differences may be attributed to variations in initial spacing, local traffic conditions, or driver response characteristics such as reaction time and sensitivity to speed changes. This comparison highlights the variability in real-world driving behavior, even under similar traffic scenarios, and highlights the difficulty of capturing these dynamics with a single model.

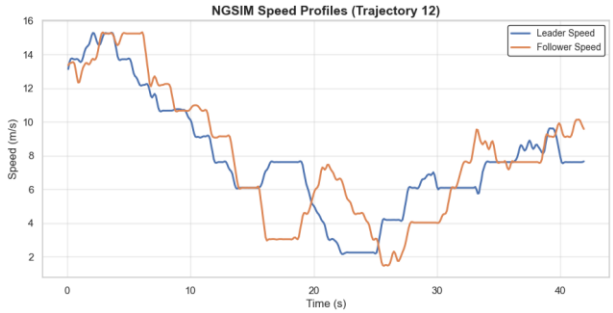

FIGURE 4: TIME-SERIES FOR TRAJECTORY 12

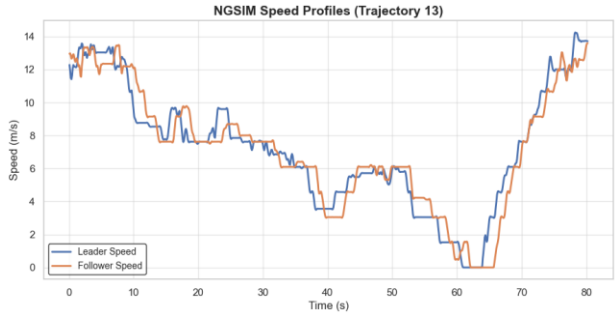

FIGURE 5: TIME-SERIES FOR TRAJECTORY 13

Headway and Relative Speed Analysis (NGSIM)

The dual-axis time-series plot in Figure 6 provides insight into interaction relationships between vehicles. Periods of decreasing headway correspond to instances where the follower vehicle is closing the gap to the leader, often accompanied by negative relative speed values.

Conversely, increases in headway are associated with positive relative speed, indicating that the follower (or leader) is increasing separation. The variability observed in both suggests that the driver continuously adjusts to maintain safe following distances.

This relationship is critical in understanding collision risk and the effectiveness of driver response. It also has direct applications in the construction of adaptive cruise control systems and automated vehicle algorithms.

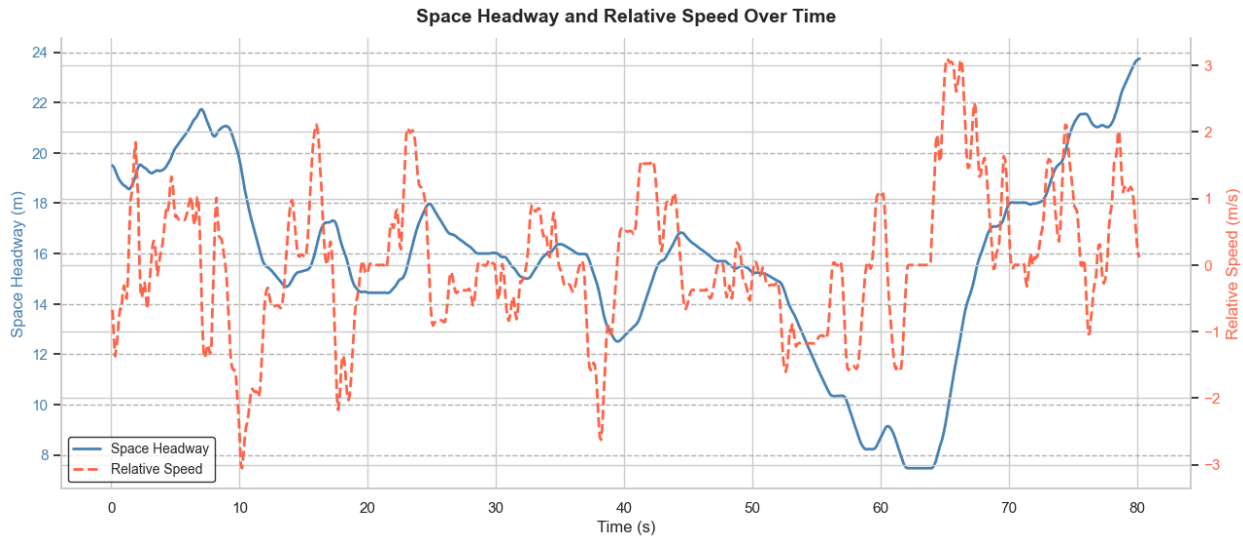

FIGURE 6: HEADWAY AND RELATIVE SPEED DUAL AXIS PLOT

Intelligent Driver Model Simulation

The IDM simulation was used to model follower vehicle behavior and compare it with observed trajectories. The simulated speed and position shown in Figure 7 closely match the observed values, indicating that the model effectively captures car-following dynamics.

The calculated performance relationships were:

- Speed RMSE = 0.728 m/s
- Position RMSE = 4.081 meters

These relatively low error values indicate a clear agreement between the model and recorded data, particularly for position. The model reliably replicates acceleration, deceleration, and general trajectory trends over time.

However, some deviations are visible during periods of rapid speed change, where the simulated response appears smoother than the observed behavior. This is a limitation of the IDM, as it assumes consistent driver behavior and does not account for sudden or irregular reactions. The IDM fails to "predict" the way a human does. When a normal driver approaches a light, and a vehicle is staged to turn in front of them, the approaching driver can slow down by "predicting" that the other driver will pull out in front. The simulation doesn't predict, it reacts. This is likely where the simulation deviates from sudden, erratic reactions which occur in reality.

Overall, the IDM provides a reliable approximation of vehicle-following behavior and is suitable for traffic simulation applications, though it may require calibration for more complex or aggressive driving conditions.

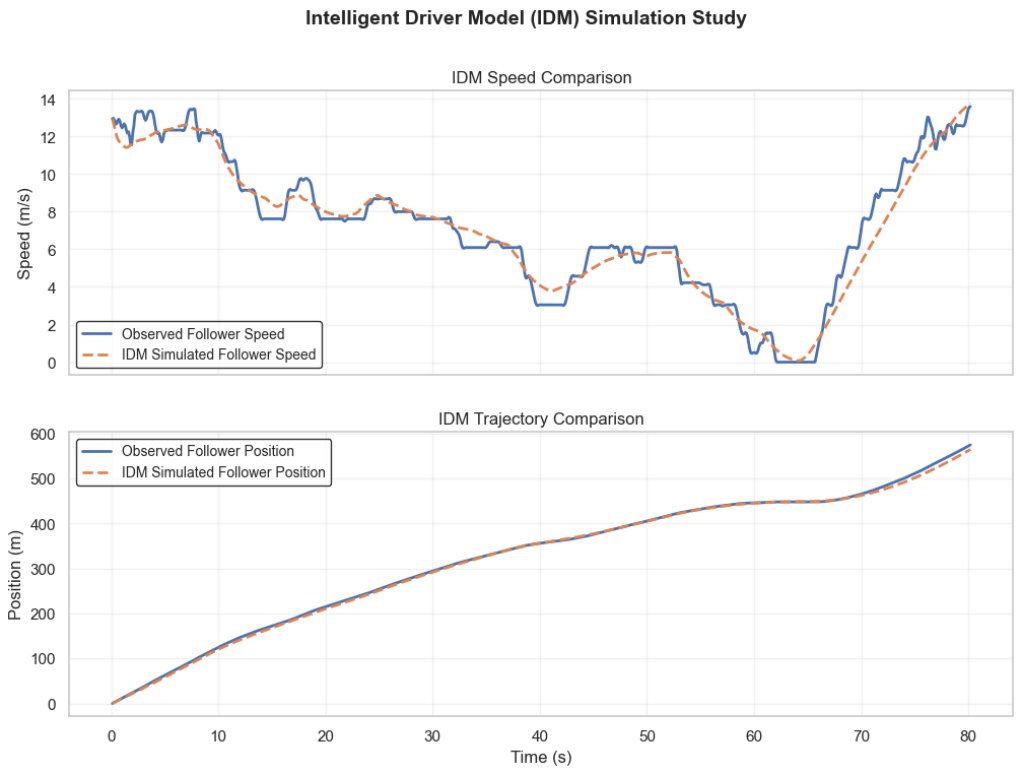

FIGURE 7: IDM SIMULATION RESULTANT COMPARISONS

Conclusion

This project proved the effectiveness of using Python to analyze and visualize transportation datasets at both large- and small-scale interactions. Overall, the time required to select variables, build and validate the code, and draw conclusions for this report was approximately 42 hours. The NHTS dataset provided insight into national travel behavior, revealing patterns in trip frequency and vehicle characteristics, while the NGSIM dataset enabled detailed analysis of vehicle interactions and driving dynamics.

The results showed clear variations in travel demand across days of the week, as well as regional differences in vehicle age that affect infrastructure planning and policy development. Also, the time-series analysis of vehicle trajectories highlighted lag responses and traffic instability, which are key contributors to congestion.

The IDM simulation further demonstrated that vehicle-following behavior can be modeled effectively using a physics-based approach, with relatively low error compared to observed data. This supports the use of simulation tools in transportation analysis and planning to avoid the substantial costs and effort requirements of conducting real-world observations.

Overall, integrating data visualization and simulation within a Python environment provides a scalable, highly capable framework for transportation engineering analysis, supporting better decision-making and future design optimization.

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